

NOTES ON



# COMMUTATIVE ALGEBRA



*Lecture Notes & Problem Discussions*

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## ABSTRACT

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These are lecture notes for a first course in Commutative Algebra, covering rings, ideals, and modules, together with the accompanying problem-set discussions worked out alongside the lectures. The standard references for this material are [\[AM69\]](#), [\[Bly90\]](#), and [\[DF04\]](#).

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## CHAPTER 1

# LECTURE NOTES

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### 1.1 Lecture 1

*Convention.* Rings are commutative with 1.

**Definition 1.1.** Let  $R, S$  be rings. A *morphism*  $\varphi$  from  $R$  to  $S$  is a map  $\varphi : R \rightarrow S$  such that

- (1)  $\varphi(r + s) = \varphi(r) + \varphi(s)$ ,
- (2)  $\varphi(rs) = \varphi(r)\varphi(s)$ ,
- (3)  $\varphi(1) = 1$ .

*Convention.* A subring  $R$  must contain 1.

*Question.* What is  $R[X]$ ?

$X$  is not an element of  $R$ . Consider

$$S = \{f : \mathbb{N} \rightarrow R \mid f(r) = 0 \text{ for all but finitely many } r\},$$

with the usual addition and multiplication in sequences (Cauchy product). Then  $S$  is a ring with identity

$$1_S = (1, 0, 0, \dots, 0).$$

Define  $\varphi : R \rightarrow S$  by  $\varphi(a) = (a, 0, \dots, 0)$ . Then  $\varphi$  is a ring homomorphism, for sure.

Define  $X = (0, 1, 0, \dots)$  and observe  $X^2 = (0, 0, 1, 0, \dots)$ . A polynomial  $P$  over  $R$  can then be identified as an element of  $S$ , say

$$P = (P_0, P_1, \dots, P_n, 0, \dots) = \varphi(P_0) \cdot 1_S + \varphi(P_1) \cdot X + \dots + \varphi(P_n) \cdot X^n.$$

**Definition 1.2.** Let  $R$  be a ring. A non-zero, non-unit element  $p \in R$  is *prime* if

$$p \mid ab \implies p \mid a \text{ or } p \mid b$$

(include the elementwise definition: for  $a, b \in R$ ,  $a \mid b$  if  $\exists r \in R$  such that  $b = r \cdot a$ ).

**Definition 1.3.** Let  $R$  be a ring. A proper ideal  $P$  of  $R$  is called a *prime ideal* if

$$ab \in P \implies a \in P \text{ or } b \in P.$$

**Definition 1.4.** Let  $R$  be a ring.  $I \subseteq R$  is called an *ideal* if there exists a ring  $S$  and a morphism  $\varphi : R \rightarrow S$  such that  $I = \ker(\varphi)$ .

**Remark 1.5.** This immediately gives the elementwise definition of ideals.

$$\mathcal{N}(R) = \{\text{all nilpotent elements of } R\} \rightsquigarrow \text{the nil-radical of } R.$$

**Theorem 1.6.**

$$\mathcal{N}(R) = \bigcap_{P \in \text{Spec } R} P, \quad \text{where } \text{Spec } R = \text{all prime ideals of } R.$$

*Proof.* Let  $x \in \mathcal{N}(R)$ . Then for any  $P \in \text{Spec } R$ , there exists  $n_P \in \mathbb{N}$  such that

$$x^{n_P} = 0 \implies 0 \in P \implies x^{n_P} \in P \implies x \in P.$$

Now let  $x \notin \mathcal{N}(R)$ , and define

$$S = \{I \subseteq R \text{ ideal} \mid x^n \notin I \text{ for all } n\}.$$

$S \neq \emptyset$ , as  $\langle 0 \rangle \in S$ , and  $S$  is a poset (ordered by inclusion). By Zorn's Lemma, let  $M$  be a maximal element of  $S$ .

**Claim.**  $M$  is a prime ideal.

*Proof of claim.* Let  $a \notin M$ ,  $b \notin M$ . Now,

$$M + \langle a \rangle, M + \langle b \rangle \notin S \implies \exists m, n \in \mathbb{N} \text{ such that}$$

$$x^m \in M + \langle a \rangle \text{ and } x^n \in M + \langle b \rangle \implies x^{m+n} \in M + \langle ab \rangle.$$

If  $ab \in M$ , then  $x^{m+n} \in M$ , a contradiction. Hence  $ab \notin M$ , so  $M$  is prime. □

Since  $x \notin M$  and  $M \in \text{Spec } R$ , we conclude  $x \notin \bigcap_{P \in \text{Spec } R} P$ , completing the proof. □

*Question.* What are the units of  $R[X]$ ?

If  $R$  is an integral domain, then they are precisely the units of  $R$ .

**Proposition 1.7.**  $f(X) = a_n X^n + \cdots + a_1 X + a_0$  is a unit if  $a_0$  is a unit and the  $a_i$ 's are nilpotent.

*Proof.* Let  $P \in \text{Spec } R$ . We have a natural morphism  $\varphi : R[X] \rightarrow (R/P)[X]$  given by

$$\varphi(b_m X^m + \cdots + b_1 X + b_0) = \overline{b_m} X^m + \cdots + \overline{b_1} X + \overline{b_0}, \quad (\overline{b_i} = b_i \bmod P).$$

Then  $\varphi(f)$  is a unit in  $(R/P)[X]$ , and since  $R/P$  is an integral domain, this forces

$$\overline{a_i} \equiv \overline{0} \pmod{P} \text{ for all } i \in [n], \quad \text{and} \quad \overline{a_0} \text{ is a unit of } R/P.$$

That is,  $a_0 \notin P$  and  $a_1, \dots, a_n \in P$ . As  $P \in \text{Spec } R$  was arbitrary,

$$a_0 \notin P \text{ and } a_i \in P \quad \forall i \in [n], \quad \text{for all } P \in \text{Spec } R.$$

The other direction is trivial. □

## 1.2 Lecture 2

**Definition 1.8.** Let  $R$  be a ring and let  $(M, +)$  be an abelian group. A *scalar multiplication*

$$R \times M \longrightarrow M, \quad (r, m) \longmapsto rm,$$

makes  $M$  into an  $R$ -module if the following hold for all  $r, s \in R$  and  $m, m_1, m_2 \in M$ :

- (i)  $(r + s)(m) = rm + sm$ ,
- (ii)  $r(m_1 + m_2) = rm_1 + rm_2$ ,
- (iii)  $r(sm) = rsm = s(rm)$ ,
- (iv)  $1 \cdot m = m$ .

**Definition 1.9.** If  $M, N$  are  $R$ -modules, a map  $\varphi : M \rightarrow N$  is called an  $R$ -module morphism if

- (i)  $\varphi(m_1 + m_2) = \varphi(m_1) + \varphi(m_2)$ ,
- (ii)  $\varphi(rm) = r\varphi(m)$ .

**Definition 1.10.** Let  $R$  be a ring and  $S$  a nonempty set. A *free  $R$ -module on  $S$*  is a pair  $(F, f)$ , where  $F$  is an  $R$ -module and  $f : S \rightarrow F$  is a set-theoretic map, such that: for any  $R$ -module  $M$  and any set-theoretic map  $g : S \rightarrow M$ , there exists a unique  $R$ -module morphism  $h : F \rightarrow M$

such that

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$

commutes, i.e.  $h \circ f = g$ .

**Proposition 1.11.** Let  $(F, f)$  be a free  $R$ -module on  $S$ . Then:

- (1)  $f$  is injective;
- (2)  $\text{Im}(f)$  generates  $F$  as an  $R$ -module.

*Proof of (1).* Let  $x, y \in S$  with  $x \neq y$ . **Goal:**  $f(x) \neq f(y)$ .

Take any  $R$ -module  $M$  with at least two elements, and define  $g : S \rightarrow M$  such that  $g(x) \neq g(y)$ . By definition, there exists a (unique) morphism  $h : F \rightarrow M$  such that  $h \circ f = g$ . Then

$$(h \circ f)(x) = g(x) \neq g(y) = (h \circ f)(y) \implies f(x) \neq f(y).$$

□

**Definition 1.12.** Let  $M$  be an  $R$ -module and  $\emptyset \neq A \subseteq M$ . The *submodule of  $M$  generated by  $A$*  is, equivalently:

- (1) the smallest submodule of  $M$  containing  $A$ ;
- (2) the intersection of all submodules of  $M$  containing  $A$ ;
- (3) the set of all finite  $R$ -linear combinations of elements of  $A$ .

*Proof of (2).* Let  $A$  be the submodule of  $F$  generated by  $\text{Im}(f)$ . Since  $f(s) \in \text{Im}(f) \subseteq A$  for every  $s \in S$ ,  $f$  factors as  $f = i_A \circ f^*$  for a set-theoretic map  $f^* : S \rightarrow A$  and the inclusion  $i_A : A \hookrightarrow F$ :

$$\begin{array}{ccc} S & \xrightarrow{f^*} & A \\ & \searrow f & \downarrow i_A \\ & & F \end{array}$$

As  $(F, f)$  is free on  $S$  and  $A$  is an  $R$ -module, there is a unique morphism  $h : F \rightarrow A$  such that  $h \circ f = f^*$ :

$$\begin{array}{ccc} S & \xrightarrow{f^*} & A \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$



It is enough to show  $i_A$  is onto. We compute

$$(i_A \circ h) \circ f = i_A \circ (h \circ f) = i_A \circ f^* = f.$$

So  $i_A \circ h$  and  $\text{id}_F$  both satisfy  $(-) \circ f = f$ ; by uniqueness in the definition of freeness (applied with  $M = F, g = f$ ),  $i_A \circ h = \text{id}_F$ . Hence  $i_A$  is onto, i.e.  $A = F$ , so  $\text{Im}(f)$  generates  $F$ .  $\square$

**Theorem 1.13: Uniqueness of free modules.** Let  $(F, f)$  be a free  $R$ -module on a set  $S$ . Then  $(F', f')$  is also a free  $R$ -module on  $S$  if and only if there exists a unique  $R$ -module isomorphism  $j : F \rightarrow F'$  such that  $j \circ f = f'$ .

*Proof.* Suppose  $(F', f')$  is also free on  $S$ . Since  $(F, f)$  is free (applied to  $M = F', g = f'$ ), there is a morphism  $h : F \rightarrow F'$  with  $h \circ f = f'$ :

$$\begin{array}{ccc} S & \xrightarrow{f'} & F' \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$

Likewise, since  $(F', f')$  is free (applied to  $M = F, g = f$ ), there is a morphism  $h' : F' \rightarrow F$  with  $h' \circ f' = f$ :

$$\begin{array}{ccc} S & \xrightarrow{f} & F \\ f' \downarrow & \nearrow h' & \\ F' & & \end{array}$$

Then  $(h' \circ h) \circ f = h' \circ (h \circ f) = h' \circ f' = f$ . So  $h' \circ h$  and  $\text{id}_F$  both satisfy  $(-) \circ f = f$ ; by uniqueness (with  $M = F, g = f$ ),  $h' \circ h = \text{id}_F$ . Similarly  $h \circ h' = \text{id}_{F'}$ . So  $j := h$  is an isomorphism with  $j \circ f = f'$ .

**Converse.** Suppose there is an isomorphism  $j : F \rightarrow F'$  with  $j \circ f = f'$ . To show  $(F', f')$  is free on  $S$ , take any  $R$ -module  $M$  and set-theoretic map  $g : S \rightarrow M$ ; we must produce a unique  $h' : F' \rightarrow M$  with  $h' \circ f' = g$ :

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f' \downarrow & \nearrow h' & \\ F' & & \end{array}$$

Since  $(F, f)$  is free, there is a unique  $h : F \rightarrow M$  with  $h \circ f = g$ . From  $j \circ f = f'$  we get  $j^{-1} \circ f' = f$ , so

$$(h \circ j^{-1}) \circ f' = h \circ (j^{-1} \circ f') = h \circ f = g,$$

i.e.  $h' := h \circ j^{-1}$  works:

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f' \downarrow & \nearrow f & \uparrow h \\ F' & \xrightarrow{j^{-1}} & F \end{array}$$

**Uniqueness.** Suppose  $t : F' \rightarrow M$  also satisfies  $t \circ f' = g$ . Then  $t \circ j : F \rightarrow M$  satisfies  $(t \circ$

$j) \circ f = t \circ (j \circ f) = t \circ f' = g$ , so by uniqueness of  $h$  (from freeness of  $(F, f)$ ),  $t \circ j = h$ , hence  $t = h \circ j^{-1} = h'$ .  $\square$

**Theorem 1.14: Existence of free  $R$ -modules.** For any ring  $R$  and nonempty set  $S$ , a free  $R$ -module on  $S$  exists.

*Proof.* Define

$$F = \{\theta : S \rightarrow R \mid \theta(s) = 0 \text{ for almost all } s \in S\}.$$

For  $\theta_1, \theta_2 \in F$  and  $\lambda \in R$ , define

$$(\theta_1 + \theta_2)(s) = \theta_1(s) + \theta_2(s), \quad (\lambda\theta)(s) = \lambda\theta(s).$$

One checks that  $F$  is an  $R$ -module. Define  $f : S \rightarrow F$  by  $f(s) = \delta_s$ , where  $\delta_s(t) = 1$  if  $t = s$  and 0 otherwise.

**Existence of  $h$ .** Let  $M$  be an  $R$ -module and  $g : S \rightarrow M$  a set-theoretic map. Define  $h : F \rightarrow M$  by

$$h(\theta) = \sum_{s \in S} \theta(s) g(s),$$

a finite sum for every  $\theta \in F$ . For any  $t \in S$ ,

$$(h \circ f)(t) = h(f(t)) = h(\delta_t) = \delta_t(t) g(t) = g(t),$$

so  $h \circ f = g$ :

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$

**Uniqueness of  $h$ .** For  $\theta \in F$  and  $t \in S$ ,

$$\theta(t) = \theta(t) \cdot 1_R = \sum_{s \in S} \theta(s) \delta_s(t) = \left( \sum_{s \in S} \theta(s) \delta_s \right) (t) \quad \forall t \in S,$$

so  $\theta = \sum_{s \in S} \theta(s) \delta_s$ . Now let  $h' : F \rightarrow M$  be any  $R$ -module morphism with  $h' \circ f = g$ . Then for  $\theta \in F$ ,

$$h'(\theta) = h' \left( \sum_{s \in S} \theta(s) \delta_s \right) = \sum_{s \in S} \theta(s) h'(\delta_s) = \sum_{s \in S} \theta(s) h'(f(s)) = \sum_{s \in S} \theta(s) g(s) = h(\theta).$$

Hence  $h = h'$ .  $\square$

### 1.3 Lecture 3

**Theorem 1.15.** Let  $R$  be a ring such that every prime ideal is principal. Then every ideal is principal.

*Proof.* Let  $S = \{\text{all non-principal ideals of } R\}$ . For the sake of contradiction (FTSOC), suppose  $S \neq \emptyset$ . By Zorn's Lemma, let  $I$  be a maximal element of  $S$ .

Since  $I$  is not principal,  $I$  is not prime by hypothesis, so there exist  $a, b \in R$  with  $a \notin I, b \notin I$ , but  $ab \in I$ .

Let  $J = \langle a \rangle + I \supsetneq I$ ; since  $I$  is maximal among non-principal ideals,  $J \notin S$ , so  $J$  is principal, say  $J = \langle c \rangle$ .

Also let  $K = \{r \in R \mid ra \in I\} \supsetneq I$  (since  $b \in K \setminus I$ ); again  $K \notin S$ , so  $K = \langle d \rangle$  for some  $d$ .

$I \subseteq \langle cd \rangle$ . Let  $x \in I$ . Since  $I \subseteq J = \langle c \rangle$ , there is  $\lambda \in R$  with  $\lambda c = x$ . Now  $\lambda a \in I$ , so  $\lambda \in K = \langle d \rangle$ , and hence  $x \in \langle cd \rangle$ . Thus  $I \subseteq \langle cd \rangle$ .

$\langle cd \rangle \subseteq I$ . There exist  $u \in R, v \in I$  such that  $c = ua + v$  (since  $c$  generates  $J = \langle a \rangle + I$ ). Also  $da \in I$  (since  $d$  generates  $K$ ). Then

$$cd = (ua + v)d = u(ad) + vd \in I,$$

since  $ad = da \in I$  and  $v \in I$ . Hence  $\langle cd \rangle \subseteq I$ .

Together,  $I = \langle cd \rangle$ , contradicting  $I \in S$ . So  $S = \emptyset$ , i.e. every ideal of  $R$  is principal.  $\square$

**Theorem 1.16.** Let  $R$  be a ring such that every prime ideal is finitely generated. Then every ideal of  $R$  is finitely generated.

*Proof.* Let  $A = \{I \text{ ideal of } R \mid I \text{ is not finitely generated}\}$ . By Zorn's Lemma, let  $M$  be a maximal element of  $A$ .

**Claim:  $M$  is a maximal ideal.** Suppose not, so there is an ideal  $N$  with  $M \subsetneq N \subsetneq R$ . Since  $N$  is maximal (hence prime), it is finitely generated by hypothesis: there exist  $a_1, \dots, a_n \in R$  such that  $N = \langle a_1, \dots, a_n \rangle$ .

Since  $M \subsetneq N$ , some  $a_i \notin M$ . Then  $M + \langle a_i \rangle \supsetneq M$ , so by maximality of  $M$  in  $A$ ,  $M + \langle a_i \rangle \notin A$ ; hence there exist  $b_1, \dots, b_k \in R$  such that

$$M + \langle a_i \rangle = \langle b_1, \dots, b_k \rangle.$$

$\square$

**Lemma 1.17.** Let  $I, J_1, J_2$  be ideals of  $R$  such that  $I \subseteq J_1 \cup J_2$ . Then either  $I \subseteq J_1$  or  $I \subseteq J_2$ .

*Proof.* FTSOC, suppose  $I \setminus J_1 \neq \emptyset$  and  $I \setminus J_2 \neq \emptyset$ . Let  $x \in I \setminus J_1$  and  $y \in I \setminus J_2$ ; since  $I \subseteq J_1 \cup J_2$ , this forces  $x \in J_2$  and  $y \in J_1$ . Also  $x + y \in I \subseteq J_1 \cup J_2$ , so  $x + y \in J_1$  or  $x + y \in J_2$ ; WLOG  $x + y \in J_1$ . Since  $y \in J_1$ ,  $x = (x + y) - y \in J_1$ , contradicting  $x \notin J_1$ .  $\square$

**Theorem 1.18: Prime Avoidance.** Let  $I, J_1, \dots, J_n$  be ideals of  $R$  such that

- (1)  $I \subseteq J_1 \cup \dots \cup J_n$ ,
- (2)  $J_3, \dots, J_n \in \text{Spec } R$ .

Then  $I \subseteq J_i$  for some  $i$ .

*Proof.* We induct on  $n$ . The case  $n = 2$  needs no primality hypothesis and is the lemma above; assume  $n \geq 3$  and the result for any collection of  $n - 1$  such ideals.

We prove the contrapositive: if  $I \not\subseteq J_i$  for every  $i$ , then  $I \not\subseteq \bigcup_{i=1}^n J_i$ .

Fix  $i$ . Apply the induction hypothesis to  $I$  together with the  $n - 1$  ideals  $\{J_j\}_{j \neq i}$  (which still has all but at most two members prime): if  $I \subseteq \bigcup_{j \neq i} J_j$ , the induction hypothesis would give  $I \subseteq J_j$  for some  $j \neq i$ , contradicting our assumption. So  $I \not\subseteq \bigcup_{j \neq i} J_j$ , and we may choose

$$z_i \in I \setminus \bigcup_{j \neq i} J_j.$$

Since  $I \subseteq \bigcup_{i=1}^n J_i$  by hypothesis (1), and  $z_i \notin J_j$  for every  $j \neq i$ , we must have  $z_i \in J_i$ , for every  $i$ .

Set

$$z := z_1 z_2 \cdots z_{n-1} + z_n \in I$$

(a sum of an element of  $I$  and a product of elements of  $I$ , hence itself in  $I$ ). We show  $z \notin J_i$  for every  $i$ , which gives  $I \not\subseteq \bigcup_{i=1}^n J_i$ , as desired.

- If  $i \in \{1, \dots, n - 1\}$ : since  $z_i \in J_i$  is one of the factors of  $z_1 \cdots z_{n-1}$ , this product lies in  $J_i$ . If  $z \in J_i$ , then  $z_n = z - z_1 \cdots z_{n-1} \in J_i$ , contradicting  $z_n \notin J_i$  (as  $i \neq n$ ).
- If  $i = n$ : if  $z \in J_n$ , then since  $z_n \in J_n$  as well,  $z_1 \cdots z_{n-1} = z - z_n \in J_n$ . As  $J_n$  is prime, some factor  $z_k \in J_n$  for a  $k \in \{1, \dots, n - 1\}$ , contradicting  $z_k \notin J_n$  (as  $k \neq n$ ).

This completes the induction. □

**Proposition 1.19.** An element of  $\mathbb{Q}(i)$  lies in  $\mathbb{Z}[i]$  if and only if it is the root of a monic polynomial over  $\mathbb{Z}$ .

*Proof.* Let  $a + bi \in \mathbb{Q}(i)$ . Then  $a + bi$  is a root of

$$P(X) = X^2 - 2aX + (a^2 + b^2) \in \mathbb{Q}[X].$$

**Claim:**  $2a \in \mathbb{Z}$  and  $a^2 + b^2 \in \mathbb{Z}$  together imply  $a, b \in \mathbb{Z}$ .

Since  $2a \in \mathbb{Z}$ ,  $(2a)^2 \in \mathbb{Z}$ . Since  $a^2 + b^2 \in \mathbb{Z}$ ,  $4(a^2 + b^2) = (2a)^2 + (2b)^2 \in \mathbb{Z}$ , so  $(2b)^2 = 4(a^2 + b^2) - (2a)^2 \in \mathbb{Z}$ . As  $2b \in \mathbb{Q}$  has integer square,  $2b \in \mathbb{Z}$ .

Now  $(2a)^2 + (2b)^2 = 4(a^2 + b^2) \equiv 0 \pmod{4}$ . Since the square of any integer is  $\equiv 0 \pmod{4}$  if it

is even, and  $\equiv 1 \pmod{4}$  if it is odd, the only way for the sum to be  $\equiv 0 \pmod{4}$  is

$$(2a)^2 \equiv (2b)^2 \equiv 0 \pmod{4},$$

forcing  $2a, 2b$  to both be even, i.e.  $a, b \in \mathbb{Z}$ .

Since  $a + bi \in \mathbb{Z}[i]$  exactly when  $a, b \in \mathbb{Z}$ , and  $a + bi$  is a root of a monic polynomial over  $\mathbb{Z}$  exactly when  $2a, a^2 + b^2 \in \mathbb{Z}$  (the coefficients of  $P(X)$ ), the claim gives the equivalence.  $\square$

**Definition 1.20.** Let  $R \hookrightarrow S$  be rings. An element  $\alpha \in S$  is called *integral* over  $R$  if  $\alpha$  is a root of a monic polynomial

$$X^n + a_{n-1}X^{n-1} + \cdots + a_0$$

over  $R$  (i.e. with  $a_0, \dots, a_{n-1} \in R$ ).

## 1.4 Lecture 4

**Definition 1.21.** Let  $R$  be a subring of  $S$ , and let  $U \subseteq S$  be a subset. Define

$$R[U] := \text{the smallest subring of } S \text{ containing } R \text{ and } U.$$

**Remark 1.22.** For  $U, V \subseteq S$ ,

$$R[U][V] = R[U \cup V] = R[V][U].$$

In particular, if  $V = \{u_1, \dots, u_n\}$ , then

$$R[u_1, \dots, u_n] = R[u_1][u_2] \cdots [u_n].$$

**Proposition 1.23.**

$$R[u] = \{a_0 + a_1u + \cdots + a_nu^n \mid a_i \in R, n \in \mathbb{N}_{\geq 0}\}.$$

*Proof.* Let  $T = \{a_0 + a_1u + \cdots + a_nu^n \mid a_i \in R, n \in \mathbb{N}_{\geq 0}\}$ .

$T$  is a **subring** of  $S$ . It contains 1 (take  $n = 0, a_0 = 1$ ). It is closed under addition: given two such expressions, pad the shorter one with zero coefficients to a common degree and add coefficientwise, which stays in  $R$  since  $R$  is a ring. It is closed under multiplication: expanding a product of two such expressions and collecting powers of  $u$ , each resulting coefficient is a finite sum of products of elements of  $R$ , hence lies in  $R$ .

$R[u] \subseteq T$ . Since  $R \subseteq T$  (take  $n = 0$ ) and  $u \in T$  (take  $n = 1, a_0 = 0, a_1 = 1$ ), and  $T$  is a subring of  $S$  containing  $R$  and  $u$ , while  $R[u]$  is by definition the *smallest* such subring, we get  $R[u] \subseteq T$ .

$T \subseteq R[u]$ . Since  $R[u]$  is a subring containing  $R$  and  $u$ , it is closed under addition and multiplication. So for any  $a_0, \dots, a_n \in R \subseteq R[u]$ , each  $a_iu^i \in R[u]$ , and hence  $a_0 + a_1u + \cdots + a_nu^n \in R[u]$ .

Combining both containments,  $R[u] = T$ . □

**Example 1.24.** The map  $\mathbb{Z}[X] \rightarrow \mathbb{Z}[i]$ ,  $X \mapsto i$ , is a surjective ring morphism with kernel  $\langle X^2 + 1 \rangle$ .

**Proposition 1.25.** Let  $R, S$  be rings and  $\varphi : R \rightarrow S$  be a morphism. Let  $\alpha \in S$  be an element. Then  $\varphi$  can be uniquely extended to a morphism  $\tilde{\varphi} : R[X] \rightarrow S$  such that  $\tilde{\varphi}|_R = \varphi$  and  $\tilde{\varphi}(X) = \alpha$ .

*Proof sketch.* Define  $\tilde{\varphi} : R[X] \rightarrow S$  by

$$\tilde{\varphi}(a_0 + a_1X + \cdots + a_nX^n) = \varphi(a_0) + \varphi(a_1)\alpha + \cdots + \varphi(a_n)\alpha^n,$$

and  $\tilde{\varphi}(A \cdot B) = \tilde{\varphi}(A) \cdot \tilde{\varphi}(B)$ . □

Let  $R$  be a subring of  $S$ , and let  $u \in S$ . Then there is a unique morphism  $\varphi : R[X] \rightarrow S$  such that  $\varphi|_R = \text{id}_R$  and  $\varphi(X) = u$  (apply the above with  $\varphi = \text{id}_R$ ,  $\alpha = u$ ). Then  $\text{Im}(\varphi) = R[u]$ ; letting  $I = \ker \varphi$ ,

$$R[X]/I \cong R[u].$$

*Notation.*  $I \cap R = \langle 0 \rangle$ .

Conversely, let  $I$  be an ideal of  $R[X]$  such that  $I \cap R = \langle 0 \rangle$ . We have the natural projection map

$$R[X] \longrightarrow R[X]/I.$$

Since  $I \cap R = 0$ ,  $R \hookrightarrow R[X]/I$ , so we can identify  $a \in R$  with its image  $a + I$ .

*Question.* What is the image of  $R[X]$  in  $R[X]/I$ ? (It will be determined by  $\text{Im}(R)$  and that of  $X$ .)

Under the projection,  $X \mapsto X + I$ . Writing  $u = X + I$ ,

$$R[X]/I \cong R[u].$$

**Definition 1.26.** Let  $R \hookrightarrow S$ .  $S$  is said to be an *integral extension* of  $R$  if each  $\alpha \in S$  is integral over  $R$ .

*Notation.* (Recall.) Let  $R$  be a commutative ring with 1, and let  $A \in M_n(R)$ . Then

$$A \cdot \text{adj}(A) = \text{adj}(A) \cdot A = \det(A) \cdot I_n.$$

**Theorem 1.27.** Let  $R \hookrightarrow S$  and  $b \in S$ . Then

$$b \text{ is integral over } R \iff R[b] \text{ is a finitely generated } R\text{-module}.$$

*Proof.* ( $\Rightarrow$ ) Assume  $b$  is integral over  $R$ ; then there is a monic polynomial  $f(X) \in R[X]$  such that

$f(b) = 0$ . Let  $\deg f = n \geq 1$ . Take any  $g(b) \in R[b]$ . By the division algorithm,

$$g(X) = q(X)f(X) + r(X), \quad \deg r < n,$$

so  $g(b) = r(b)$ . Hence  $R[b]$  is generated (as an  $R$ -module) by  $\{1, b, \dots, b^{n-1}\}$ .

( $\Leftarrow$ ) Conversely, let  $R[b]$  be a finitely generated  $R$ -module, say  $R[b] = \langle w_1, \dots, w_n \rangle$ . Then for each  $i \in [n]$ ,

$$bw_i = a_{i1}w_1 + \dots + a_{in}w_n, \quad a_{ij} \in R,$$

i.e.  $(bI - A)\vec{w} = 0$ , where  $A = (a_{ij})$ . Multiplying by  $\text{adj}(bI - A)$ ,

$$\det(bI - A)w_i = 0 \quad \forall i \in [n].$$

Since  $1 \in R[b] = \langle w_1, \dots, w_n \rangle$ , we get  $\det(bI - A) = 0$ , which is a monic equation in  $b$  (of degree  $n$ ). Hence  $b$  is integral over  $R$ .  $\square$

**Theorem 1.28.** Let  $R \hookrightarrow S$ , and let  $b_1, b_2, \dots, b_n \in S$ . Then each  $b_i$  is integral over  $R$  if and only if  $R[b_1, \dots, b_n]$  is a finitely generated  $R$ -module.

*Proof.* Induct on  $n$ . The case  $n = 1$  is the previous theorem.

For  $n > 1$ : assume  $b_1, \dots, b_n$  are integral over  $R$ . In particular  $b_n$  is integral over  $R$ , hence also integral over  $R[b_1, \dots, b_{n-1}]$ ; by the previous theorem (applied over the base ring  $R[b_1, \dots, b_{n-1}]$ ),  $R[b_1, \dots, b_{n-1}][b_n]$  is a finitely generated  $R[b_1, \dots, b_{n-1}]$ -module, which (combined with the induction hypothesis that  $R[b_1, \dots, b_{n-1}]$  is a finitely generated  $R$ -module) makes  $R[b_1, \dots, b_n]$  a finitely generated  $R$ -module.

The converse is the same as the  $n = 1$  case.  $\square$

**Exercise 1.29.** Let  $R \hookrightarrow S$ , and let  $a, b \in S$  be integral over  $R$ . Then  $a \pm b$  and  $ab$  are integral over  $R$ .

## 1.5 Lecture 5

**Corollary 1.30.** Let  $R \hookrightarrow S$  and  $b_1, \dots, b_n \in S$  be integral over  $R$  (equivalently,  $R[b_1, \dots, b_n]$  is a finitely generated  $R$ -module). Then every  $b \in R[b_1, \dots, b_n]$  is integral over  $R$ .

*Proof.* Since  $b \in R[b_1, \dots, b_n]$ ,

$$R[b_1, \dots, b_n] = R[b_1, \dots, b_n, b],$$

which is a finitely generated  $R$ -module. By the previous theorem,  $b$  is integral over  $R$ .  $\square$

**Proposition 1.31.** Let  $A \subseteq B \subseteq C$  be rings such that  $B$  is integral over  $A$  and  $C$  is integral over  $B$ . Then  $C$  is integral over  $A$ .

*Proof.* Let  $c \in C$ . Then  $c$  is integral over  $B$ :

$$c^n + b_{n-1}c^{n-1} + \cdots + b_0 = 0, \quad b_i \in B.$$

Let  $R := A[b_0, \dots, b_{n-1}]$ . Then  $c$  is integral over  $R$ , so  $R[c]$  is a finitely generated  $R$ -module.

Since  $b_0, \dots, b_{n-1}$  are integral over  $A$ ,  $R$  is a finitely generated  $A$ -module. Hence  $R[c]$  is a finitely generated  $A$ -module, and since  $c \in R[c]$ ,  $c$  is integral over  $A$ .  $\square$

**Definition 1.32.** Define  $\bar{R}^S := \{r \in S \mid r \text{ is integral over } R\}$ . Note that  $\bar{R}^S$  is a subring of  $S$ . This  $\bar{R}^S$  is called the *integral closure* of  $R$  in  $S$ .

**Definition 1.33.**  $R$  is said to be *integrally closed* in  $S$  if  $\bar{R}^S = R$ .

$$R \hookrightarrow \bar{R}^S \hookrightarrow S.$$

**Observation 1.34.**  $\bar{R}^S$  is integrally closed in  $S$ .

**Definition 1.35.** Let  $R$  be an integral domain with quotient field  $F = \mathbb{Q}(R)$ . We say that  $R$  is *integrally closed* if  $R$  is integrally closed in  $F$ .

**Proposition 1.36.** If  $R$  is a UFD, then  $R$  is integrally closed.

*Proof.* Let  $F = \mathbb{Q}(R)$ . Say  $r/s \in F$  with  $r, s \in R$ ,  $s \neq 0$ , is integral over  $R$ , and suppose  $r, s$  have no common prime factors. Then

$$\left(\frac{r}{s}\right)^n + a_{n-1}\left(\frac{r}{s}\right)^{n-1} + \cdots + a_1\left(\frac{r}{s}\right) + a_0 = 0, \quad a_i \in R,$$

so

$$r^n + a_{n-1}r^{n-1}s + \cdots + a_1rs^{n-1} + a_0s^n = 0.$$

Since  $s$  divides every term but the first,  $s \mid r^n$ . FTSOC, if  $s$  is not a unit, it has a prime factor  $p$ ; then  $p \mid r^n$ , and since  $p$  is prime,  $p \mid r$  – contradicting that  $r, s$  have no common prime factors. Hence  $s$  is a unit in  $R$ , so  $r/s \in R$ .  $\square$

**Example 1.37.**  $\mathbb{Z}[\sqrt{-5}]$  is integrally closed but not a UFD. (Exercise.)

**Example 1.38.**  $\mathbb{Z}[\sqrt{5}]$  is an integral domain which is not integrally closed.



*Proof.*  $\mathbb{Q}(\mathbb{Z}[\sqrt{5}]) = \mathbb{Q}(\sqrt{5})$ , and

$$\frac{1 + \sqrt{5}}{2} \in \mathbb{Q}(\sqrt{5}) \setminus \mathbb{Z}[\sqrt{5}]$$

is a root of the monic polynomial  $X^2 - X - 1 \in \mathbb{Z}[X]$ , hence integral over  $\mathbb{Z}[\sqrt{5}]$ , yet not itself an element of  $\mathbb{Z}[\sqrt{5}]$ .  $\square$

## Extensions and Contractions

Let  $\varphi : R \rightarrow S$  be a ring morphism. Then  $\varphi$  factors as

$$R \xrightarrow{\varphi} \varphi(R) \xhookrightarrow{i} S \quad \text{with} \quad R / \text{Ker } \varphi \cong \varphi(R).$$

- (1) If  $I \subseteq R$  is an ideal,  $\varphi(I)$  need *not* be an ideal of  $S$ .
- (2) If  $J \subseteq S$  is an ideal,  $\varphi^{-1}(J)$  is an ideal of  $R$ .
- (3) If  $Q \in \text{Spec } S$ , then  $\varphi^{-1}(Q) \in \text{Spec } R$ .

*Question.* Suppose  $P \in \text{Spec } R$ . Can we say that  $\langle \varphi(P) \rangle$  is prime in  $S$ ?

Consider  $\mathbb{Z} \hookrightarrow \mathbb{Z}[i]: \langle 2 \rangle \mapsto \langle 1+i \rangle \langle 1-i \rangle$ . If  $p \equiv 1 \pmod{4}$  is prime, then  $\langle p \rangle$  is *not* prime in  $\mathbb{Z}[i]$  (so the answer to the question above is no, in general).

**Exercise 1.39.** If  $p \equiv 3 \pmod{4}$ , then  $\langle p \rangle$  is a prime ideal in  $\mathbb{Z}[i]$ .

**Definition 1.40.** Let  $\varphi : R \rightarrow S$  be a ring morphism.

- For  $I$  an ideal of  $R$ , the *extension*  $I^e$  is the ideal of  $S$  generated by  $\varphi(I)$ .
- For  $J$  an ideal of  $S$ , the *contraction*  $J^c := \varphi^{-1}(J)$ , an ideal of  $R$ .

**Exercise 1.41.**

$$I \subseteq I^{ec}, \quad J \supseteq J^{ce}, \quad I^e = I^{ece}, \quad J^c = J^{cec}.$$

## Rings of Fractions

**Definition 1.42.** Let  $R$  be a ring. A subset  $S \subseteq R$  is called *multiplicative* if

- (1)  $1 \in S$ ,
- (2)  $a, b \in S \implies ab \in S$ .

**Construction 1.43.** On  $R \times S$ , define the relation  $\sim$  by

$$(a, s) \sim (b, t) \iff \exists u \in S \text{ such that } u(at - bs) = 0.$$

$\sim$  is transitive. Suppose  $(a, s) \sim (b, t)$  and  $(b, t) \sim (c, u)$ ; so  $(at - bs)v = 0$  for some  $v \in S$ , and  $(bu - ct)w = 0$  for some  $w \in S$ . Then

$$atv - bsv = 0 \rightsquigarrow uw \cdot atv - bsuw = 0, \quad buw - ctw = 0 \rightsquigarrow buwsv - ctwsv = 0.$$

Adding,

$$(au - cs)vtw = 0, \quad vtw \in S,$$

so  $(a, s) \sim (c, u)$ . □

*Notation.* The equivalence class of  $(a, s)$  is denoted  $\frac{a}{s}$ .

Let

$$S^{-1}R := \{\text{all equivalence classes}\} = \left\{ \frac{a}{s} \mid a \in R, s \in S \setminus \{0\} \right\}.$$

Define

$$\frac{a}{s} + \frac{b}{t} := \frac{at + bs}{st} \quad (\text{well-defined}), \quad \frac{a}{s} \cdot \frac{b}{t} := \frac{ab}{st} \quad (\text{well-defined}).$$

$S^{-1}R$  is a commutative ring with 1. We have a canonical ring morphism

$$\varphi : R \rightarrow S^{-1}R, \quad r \mapsto \frac{r}{1}.$$

**Remark 1.44.**  $\varphi$  is not injective in general.

**Proposition 1.45.** Let  $g : R \rightarrow A$  be a ring morphism such that  $g(S) \subseteq A^\times$  (the units of  $A$ ). Then there is a unique morphism  $h : S^{-1}R \rightarrow A$  such that

$$\begin{array}{ccc} R & \xrightarrow{g} & A \\ \varphi \downarrow & \nearrow h & \\ S^{-1}R & & \end{array}$$

commutes, i.e.  $h \circ \varphi = g$ .

*Proof.* **Uniqueness.** Suppose such an  $h$  exists. Then

$$h\left(\frac{r}{1}\right) = h(\varphi(r)) = g(r), \quad h\left(\frac{1}{s}\right) = h\left(\left(\frac{s}{1}\right)^{-1}\right) = h\left(\frac{s}{1}\right)^{-1} = g(s)^{-1},$$

so

$$h\left(\frac{r}{s}\right) = h\left(\frac{r}{1}\right)h\left(\frac{1}{s}\right) = g(r)g(s)^{-1},$$

i.e.  $h$  is uniquely determined by  $g$ .

**Existence.** Define  $h : S^{-1}R \rightarrow A$  by  $h(r/s) = g(r)g(s)^{-1}$ .

*Well-definedness.* Suppose  $r/s = r'/s'$ ; then there is  $t \in S$  with  $t(rs' - r's) = 0$ , so

$$g(t)(g(r)g(s') - g(r')g(s)) = 0.$$

Since  $g(t) \in A^\times$ ,  $g(r)g(s') = g(r')g(s)$ , i.e.  $g(r)g(s)^{-1} = g(r')g(s')^{-1}$ . □

**Example 1.46.** Let  $P \in \text{Spec } R$  and  $S = R \setminus P$ . Then  $S$  is multiplicative; write  $S^{-1}R =: R_P$ . Let

$$PR_P := \left\{ \frac{p}{s} \mid p \in P, s \in R \setminus P \right\},$$

an ideal of  $R_P$ . If  $b/t \notin PR_P$ , then  $b \notin P$ , so  $b \in R \setminus P$ , so  $b/t$  is a unit in  $R_P$ . Hence  $PR_P$  is the unique maximal ideal of  $R_P$ , so  $R_P$  is local.

**Example 1.47.** Let  $f \in R$  and  $S = \{1, f, f^2, \dots\}$ . Then  $R_f := S^{-1}R$ .

**Construction 1.48.** Let  $R$  be a ring,  $S$  a multiplicative subset of  $R$ , and  $M$  an  $R$ -module. On  $M \times S$ , define  $\sim$  by

$$(m, s) \sim (m', s') \iff \exists t \in S \text{ such that } t(ms' - m's) = 0.$$

Then  $S^{-1}M := (M \times S)/\sim$  is an  $S^{-1}R$ -module.

Let  $M, N$  be  $R$ -modules and  $\varphi : M \rightarrow N$  an  $R$ -module morphism. Define  $S^{-1}\varphi : S^{-1}M \rightarrow S^{-1}N$  by

$$\frac{m}{s} \mapsto \frac{\varphi(m)}{s}.$$

$S^{-1}\varphi$  is a well-defined  $S^{-1}R$ -module morphism.

**Proposition 1.49.**  $S^{-1}(-)$  preserves exactness: if

$$M' \xrightarrow{\varphi} M \xrightarrow{\psi} M''$$

is exact at  $M$ , then

$$S^{-1}M' \xrightarrow{S^{-1}\varphi} S^{-1}M \xrightarrow{S^{-1}\psi} S^{-1}M''$$

is exact at  $S^{-1}M$ .

*Proof.* We have  $\psi \circ \varphi = 0$ , so  $(S^{-1}\psi) \circ (S^{-1}\varphi) = S^{-1}(\psi \circ \varphi) = 0$ . Hence  $\text{Im}(S^{-1}\varphi) \subseteq \text{Ker}(S^{-1}\psi)$ .

Conversely, let  $m/s \in \text{Ker}(S^{-1}\psi)$ . Then  $\psi(m)/s = 0$ , so there is  $t \in S$  with  $t\psi(m) = 0$ , i.e.

$\psi(tm) = 0$ , i.e.  $tm \in \text{Ker}(\psi) = \text{Im}(\varphi)$ . So there is  $m' \in M'$  with  $\varphi(m') = tm$ . Then

$$\frac{m}{s} = \frac{mt}{st} = \frac{\varphi(m')}{st} = (S^{-1}\varphi)\left(\frac{m'}{st}\right) \in \text{Im}(S^{-1}\varphi).$$

Hence  $\text{Ker}(S^{-1}\psi) \subseteq \text{Im}(S^{-1}\varphi)$ , giving equality.  $\square$

## 1.6 Lecture 6

**Theorem 1.50.** Let  $M$  be an  $R$ -module generated by  $n$  elements, and let

$$\varphi \in \text{Hom}(M, M) := \{f : M \rightarrow M \mid f \text{ is an } R\text{-module morphism}\}.$$

Let  $I$  be an ideal of  $R$  such that  $\varphi(M) \subseteq IM$ . Then there is a relation of the form

$$\varphi^n + a_{n-1}\varphi^{n-1} + \cdots + a_0 = 0, \quad a_i \in I^i.$$

*Proof.* Let  $M = Rw_1 + \cdots + Rw_n$ , so  $IM = Iw_1 + \cdots + Iw_n$ ; in particular  $\varphi(M) \subseteq IM$ . Write

$$\varphi(w_i) = \sum_{j=1}^n a_{ij}w_j, \quad a_{ij} \in I \text{ for all } i,$$

i.e.

$$\sum_{j=1}^n (\varphi \delta_{ij} - a_{ij})w_j = 0.$$

The coefficient matrix has  $(i, j)$ -entry  $\varphi \delta_{ij} - a_{ij}$ , with entries in the commutative subring of  $\text{End}(M)$  generated by  $R$  (acting by scalars) and  $\varphi$ . Let  $d = \det(\text{coefficient matrix})$ . Multiplying by the adjugate matrix,

$$d \cdot w_k = 0 \quad \forall k \in [n],$$

so  $d \cdot M = 0$ , i.e.  $d$  is the zero endomorphism of  $M$ . Expanding the determinant gives exactly a relation of the stated form.  $\square$

**Theorem 1.51: Nakayama's Lemma.** Let  $M$  be a finitely generated  $R$ -module and  $I$  an ideal of  $R$ . If  $M = IM$ , then there exists  $a \in R$  such that  $aM = 0$  and  $a \equiv 1 \pmod{I}$ . Furthermore, if  $I$  is the Jacobson radical of  $R$ , then  $M = 0$ .

*Proof.* Take  $\varphi = \text{id}_M$ . By the previous theorem, we have

$$1 + a_{n-1} + \cdots + a_0 = 0 \quad (\text{as endomorphisms of } M).$$

Define  $a := 1 + a_{n-1} + \cdots + a_0$ . Then  $aM = 0$  and  $a \equiv 1 \pmod{I}$ .

If  $I \subseteq J(R)$  (the Jacobson radical), then  $a - 1 \in J(R)$ , so  $a$  is a unit; hence  $aM = 0$  forces  $M = 0$ .

$$a^{-1}(aM) = 0. \quad \square$$

**Theorem 1.52.** Let  $M$  be a finitely generated  $R$ -module and  $I$  an ideal of  $R$  such that  $M = IM$ . Then there exists  $i \in I$  such that  $(1 + i)M = 0$ .

*Proof (Ojanguren).* Induct on the minimal number of generators  $\nu(M) = n$  of  $M$ .

If  $\nu(M) = 0$ , then  $M = 0$  and we are done.

Let  $M = Rm_1 + \cdots + Rm_n$ , and set  $M' := M/Rm_n$ . Then

$$IM' = \frac{IM + Rm_n}{Rm_n} = \frac{M + Rm_n}{Rm_n} = \frac{M}{Rm_n} = M',$$

using  $IM = M$ . Since  $\nu(M') < n$ , by induction there is  $x \in I$  such that  $(1 + x)M' = 0$ , i.e.

$$(1 + x) \frac{M}{Rm_n} = 0 \implies (1 + x)M \subseteq Rm_n.$$

Then

$$(1 + x)M = (1 + x)IM = I(1 + x)M \subseteq I \cdot Rm_n = Im_n,$$

using  $(1 + x)M \subseteq Rm_n$  in the last containment. In particular  $(1 + x)m_n \in Im_n$ , so  $(1 + x)m_n = ym_n$  for some  $y \in I$ , giving

$$(1 + x - y)m_n = 0.$$

Now for any  $\mu \in M$ ,  $(1 + x)\mu = rm_n$  for some  $r \in R$  (since  $(1 + x)M \subseteq Rm_n$ ), so

$$(1 + x - y)(1 + x)\mu = r(1 + x - y)m_n = 0.$$

Hence  $(1 + x)(1 + x - y)M = 0$ . Expanding,  $(1 + x)(1 + x - y) = 1 + i$  where  $i := x + (x - y) + x(x - y) \in I$ . So  $(1 + i)M = 0$ .  $\square$

**Definition 1.53.** Let  $M, N$  be  $R$ -modules. The *tensor product* of  $M, N$  is a pair  $(T, \theta)$  such that  $T$  is an  $R$ -module,  $\theta : M \times N \rightarrow T$  is bilinear, and for every bilinear map  $f : M \times N \rightarrow P$ , there is a unique linear map  $h : T \rightarrow P$  such that

$$\begin{array}{ccc} M \times N & \xrightarrow{f} & P \\ \theta \downarrow & \nearrow h & \\ T & & \end{array}$$

commutes, i.e.  $h \circ \theta = f$ .

Recall the canonical morphism  $R \rightarrow S^{-1}R$ ,  $r \mapsto r/1$ , and let  $I$  be an ideal of  $R$ . An element of the

extended ideal  $I^e$  is a finite sum

$$\frac{r_1}{s_1} \cdot \frac{x_1}{1} + \cdots + \frac{r_n}{s_n} \cdot \frac{x_n}{1} = \frac{s'_1 r_1 x_1 + \cdots + s'_n r_n x_n}{s_1 \cdots s_n},$$

where  $r_i \in R$ ,  $s_i \in S$ ,  $x_i \in I$  (writing  $s'_i$  for the product of all  $s_j$  with  $j \neq i$ ).

**Proposition 1.54.**

- (1) Every ideal of  $S^{-1}R$  is an extended ideal.
- (2) The prime ideals of  $S^{-1}R$  are in one-to-one correspondence,  $P \leftrightarrow S^{-1}P$ , with the prime ideals  $P$  of  $R$  such that  $S \cap P = \emptyset$ .

*Proof.* See [AM69]. □

### Local Properties of Modules

**Proposition 1.55.** Let  $M$  be an  $R$ -module. Then the following are equivalent:

- (1)  $M = 0$ ;
- (2)  $M_P = 0$  for all  $P \in \text{Spec}(R)$ ;
- (3)  $M_{\mathfrak{m}} = 0$  for all  $\mathfrak{m} \in \text{Max}(R)$ .

(Here for  $P \in \text{Spec}(R)$ , writing  $S = R \setminus P$ , we set  $M_P := S^{-1}M$ .)

*Proof.* (1)  $\Rightarrow$  (2)  $\Rightarrow$  (3) trivially.

(3)  $\Rightarrow$  (1): Let  $M \neq 0$ ; then there is  $x \in M$  with  $x \neq 0$ . Let  $I = \text{Ann}(x) = \{r \in R \mid rx = 0\}$ , a proper ideal of  $R$ . Let  $\mathfrak{m}$  be a maximal ideal with  $I \subseteq \mathfrak{m}$ .

Consider  $x/1 \in M_{\mathfrak{m}}$ . Since  $M_{\mathfrak{m}} = 0$ ,  $x/1 = 0$ , so there is  $t \in R \setminus \mathfrak{m}$  with  $tx = 0$ . Then  $t \in I \subseteq \mathfrak{m}$  – contradicting  $t \notin \mathfrak{m}$ . □

**Exercise 1.56.** Let  $\varphi \in \text{Hom}(M, N)$ . TFAE:

- (1)  $\varphi$  is injective (resp. surjective);
- (2)  $\varphi_P : M_P \rightarrow N_P$  is injective (resp. surjective) for all  $P \in \text{Spec}(R)$ ;
- (3)  $\varphi_{\mathfrak{m}} : M_{\mathfrak{m}} \rightarrow N_{\mathfrak{m}}$  is injective (resp. surjective) for all  $\mathfrak{m} \in \text{Max}(R)$ .

**Proposition 1.57.** Let  $R \hookrightarrow S$  be rings with  $S$  integral over  $R$ .

- (1) If  $J$  is an ideal of  $S$ , then  $R/(J \cap R) \hookrightarrow S/J$  is integral.
- (2) If  $T$  is a multiplicative subset of  $R$ , then  $T^{-1}R \hookrightarrow T^{-1}S$  is integral.

*Proof.* (1) is straightforward (reduce the monic integral equation modulo  $J$ ).

(2) Let  $x/t \in T^{-1}S$  ( $x \in S, t \in T$ ). Since  $x$  is integral over  $R$ ,

$$x^n + a_1x^{n-1} + \cdots + a_n = 0 \quad \text{for some } a_i \in R.$$

Dividing by  $t^n$ ,

$$\left(\frac{x}{t}\right)^n + \frac{a_1}{t} \left(\frac{x}{t}\right)^{n-1} + \cdots + \frac{a_n}{t^n} = 0,$$

so  $x/t$  is integral over  $T^{-1}R$ . □

**Proposition 1.58.** Let  $R \hookrightarrow S$  be an integral extension, with  $R, S$  both integral domains. Then  $R$  is a field if and only if  $S$  is a field.

*Proof.* ( $\Rightarrow$ ) Let  $R$  be a field and  $y \in S \setminus \{0\}$ . Since  $y$  is integral over  $R$ , take a monic relation of minimal degree,

$$y^n + a_1y^{n-1} + \cdots + a_n = 0, \quad a_i \in R.$$

If  $a_n = 0$ , then  $y(y^{n-1} + a_1y^{n-2} + \cdots + a_{n-1}) = 0$ ; since  $S$  is a domain and  $y \neq 0$ , this gives a relation of degree  $n - 1$ , contradicting minimality. So  $a_n \neq 0$ , and as  $R$  is a field,  $a_n$  is a unit. Then

$$y(y^{n-1} + \cdots + a_{n-1}) = -a_n,$$

and since  $-a_n$  is a unit,  $y$  is a unit.

( $\Leftarrow$ ) Let  $S$  be a field. Take any  $x \in R \setminus \{0\}$ . Then  $x^{-1} \in S$  is integral over  $R$ :

$$x^{-m} + a_1x^{-(m-1)} + \cdots + a_m = 0, \quad a_i \in R,$$

so

$$x^{-1} = -(a_1 + a_2x + \cdots + a_mx^{m-1}) \in R.$$

□

## 1.7 Lecture 7

**Example 1.59.** Let  $k$  be a field,  $t$  an indeterminate, and  $A = k[t^2, t^3] \subseteq B = k[t]$ . Since  $A$  is a subring of the integral domain  $B$ ,  $A$  is an integral domain. The fraction field of both  $A$  and  $B$  is  $k(t)$ , and since  $B$  is a UFD,  $B$  is integrally closed in  $k(t)$ .

*Question.* Is  $A$  integrally closed in  $k(t)$ ?

No:  $t$  is integral over  $A$ , being a root of  $X^2 - t^2 \in A[X]$ , but  $t \notin A$ . So  $A$  is not integrally closed.

*Question.* What is the integral closure of  $A$  in  $k(t)$ ?

It is  $B$ , since  $B$  is integrally closed in  $k(t)$ .

**Example 1.60.** Similarly,

$$\begin{array}{ccc} B & \hookrightarrow & \mathbb{Q}(i) \\ \uparrow & & \uparrow \\ \mathbb{Z} & \hookrightarrow & \mathbb{Q} \end{array}$$

where  $B$  is the integral closure of  $\mathbb{Z}$  (an integrally closed domain) in  $\mathbb{Q}(i)$ . One checks  $B = \mathbb{Z}[i]$ :  $\min(i, \mathbb{Q}) = X^2 + 1 \in \mathbb{Z}[X]$ .

**Theorem 1.61.** Let  $A$  be an integrally closed domain with  $K = \mathbb{Q}(A)$ , and let  $L$  be an algebraic extension of  $K$ . Then  $\alpha \in L$  is integral over  $A$  if and only if  $\min(\alpha, K) \in A[X]$ .

*Proof.* ( $\Leftarrow$ ) is trivial: if  $f(X) = X^n + a_1X^{n-1} + \cdots + a_n = \min(\alpha, K) \in A[X]$ , then  $f(\alpha) = 0$  exhibits  $\alpha$  as integral over  $A$ .

( $\Rightarrow$ ) Let  $\alpha$  be integral over  $A$ ; we show the coefficients  $a_i$  of  $f := \min(\alpha, K)$  lie in  $A$ .

Let  $\bar{L}$  be an algebraic closure of  $L$ . In  $\bar{L}$ ,  $f$  splits as

$$f(X) = (X - \alpha_1) \cdots (X - \alpha_n),$$

where the  $\alpha_i$  are the conjugates of  $\alpha$  over  $K$ . For each  $i$  there is a  $K$ -isomorphism  $K[\alpha] \cong K[\alpha_i]$  fixing  $K$  and sending  $\alpha \mapsto \alpha_i$ .

Since  $\alpha$  is integral over  $A$ , there is a monic  $g(X) \in A[X]$  with  $g(\alpha) = 0$ ; as  $f = \min(\alpha, K)$ ,  $f \mid g$  in  $K[X]$ . Applying the isomorphism above,  $g(\alpha_i) = 0$  for each  $i$ , so each  $\alpha_i$  is integral over  $A$ .

Now  $a_1, \dots, a_n \in A[\alpha_1, \dots, \alpha_n]$  (they are, up to sign, the elementary symmetric functions of the  $\alpha_i$  – Newton's identities), and each  $\alpha_i$  is integral over  $A$ , so each  $a_i$  is integral over  $A$ . But  $a_i \in K$  and  $A$  is integrally closed in  $K$ , so  $a_i \in A$  for all  $i$ .  $\square$

**Exercise 1.62.**

- (1)  $k[X, Y]/\langle Y^2 - X^3 \rangle \cong k[t^2, t^3]$  (the coordinate ring of the curve  $Y^2 = X^3$ ).
- (2) Show directly that  $k[t^2, t^3]$  is not a UFD.

**Example 1.63.** Let  $A$  be a UFD with  $2 \in A^\times$ , and let  $f \in A$  be square-free. Then  $A[\sqrt{f}]$  is an integrally closed domain.

*Proof.* Since  $\langle X^2 - f \rangle \in \text{Spec}(A[X])$  (as  $f$  is square-free,  $X^2 - f$  is irreducible),

$$A[\sqrt{f}] \cong A[X]/\langle X^2 - f \rangle$$

is an integral domain.

Let  $\alpha$  be a square root of  $f$ ; we ask whether  $A[\alpha]$  is integrally closed. Let  $K := \mathbb{Q}(A)$ ; since  $A$  is a



UFD,  $A$  is integrally closed in  $K$ .

**Case  $\alpha \in K$ :** Then  $X^2 - f \in A[X]$  has root  $\alpha \in K$ , so  $\alpha$  is integral over  $A$ , hence  $\alpha \in A$  (as  $A$  is integrally closed in  $K$ ). So  $A[\alpha] = A$ , trivially integrally closed.

**Case  $\alpha \notin K$ :** Then

$$\mathbb{Q}(A[\alpha]) = K(\alpha) = K + K\alpha,$$

and every  $\theta \in K(\alpha)$  can be written uniquely as  $\theta = x + y\alpha$  with  $x, y \in K$ ; its minimal polynomial over  $K$  is

$$X^2 - 2xX + (x^2 - fy^2).$$

Suppose  $\theta$  is integral over  $A[\alpha]$ . Since  $\alpha$  is integral over  $A$  (a root of  $X^2 - f \in A[X]$ ),  $A[\alpha]$  is integral over  $A$ ; by transitivity,  $\theta$  is integral over  $A$ . By the previous theorem (as  $A$  is integrally closed in  $K$ ),

$$X^2 - 2xX + (x^2 - fy^2) \in A[X],$$

i.e.  $2x \in A$  and  $x^2 - fy^2 \in A$ . Since 2 is a unit,  $x \in A$ ; hence  $fy^2 = x^2 - (x^2 - fy^2) \in A$ .

If some prime  $p \in A$  divides the denominator of  $y$  (writing  $y \in K$  in lowest terms), then since  $fy^2 \in A$  and  $f$  is square-free, we would need  $p^2 \mid f$  – a contradiction. Hence  $y \in A$ , so  $\theta = x + y\alpha \in A[\alpha]$ .

Together with the trivial case  $A[\alpha] = A$ , this shows  $A[\alpha]$  is integrally closed.  $\square$

**Lemma 1.64.** Let  $A \hookrightarrow B$  be an integral extension. If  $P \in \text{Max}(B)$ , then  $P \cap A \in \text{Max}(A)$ . Conversely, if  $\mathfrak{p} \in \text{Max}(A)$ , then there exists  $P \in \text{Spec}(B)$  such that  $P \cap A = \mathfrak{p}$  (and  $P$  is maximal too).

*Proof.*  $P \cap A \in \text{Spec}(A)$ , and  $A/(P \cap A) \hookrightarrow B/P$  is an integral extension. Since  $B/P$  is a field,  $A/(P \cap A)$  is a field, so  $P \cap A \in \text{Max}(A)$ .

Conversely, let  $\mathfrak{p} \in \text{Max}(A)$ . It suffices to show  $\mathfrak{p}B \neq B$ : if  $\mathfrak{p}B$  is a proper ideal, there is  $P \in \text{Max}(B)$  with  $\mathfrak{p}B \subseteq P$ , so  $\mathfrak{p} \subseteq P \cap A$ ; since  $1 \notin P \cap A$  and  $\mathfrak{p}$  is maximal,  $P \cap A = \mathfrak{p}$ .

Suppose instead  $\mathfrak{p}B = B$ . Then  $1 = b_1p_1 + \cdots + b_np_n$  for some  $b_i \in B, p_i \in \mathfrak{p}$ . Let  $C = A[b_1, \dots, b_n]$ ; since each  $b_i$  is integral over  $A$ ,  $C$  is a finitely generated  $A$ -module. From  $1 = \sum b_ip_i$  (with  $b_i \in C, p_i \in \mathfrak{p}$ ),  $\mathfrak{p}C = C$ .

By Nakayama's Lemma, there is  $a \in A$  with  $a \equiv 1 \pmod{\mathfrak{p}}$  and  $aC = 0$ ; since  $1 \in C, a = a \cdot 1 = 0$ , so  $0 \equiv 1 \pmod{\mathfrak{p}}$ , i.e.  $1 \in \mathfrak{p}$  – a contradiction.  $\square$

**Theorem 1.65: Lying Over.** Let  $A \hookrightarrow B$  be an integral extension and  $\mathfrak{p} \in \text{Spec}(A)$ .

- (i) There exists  $P \in \text{Spec}(B)$  such that  $P \cap A = \mathfrak{p}$  (we say  $P$  lies over  $\mathfrak{p}$ ).
- (ii) There is no inclusion between distinct prime ideals of  $B$  lying over  $\mathfrak{p}$ : if  $P \subseteq Q$  and  $P \cap A = \mathfrak{p} = Q \cap A$ , then  $P = Q$ .

*Proof.* Localize at  $S = A \setminus \mathfrak{p}$ : we obtain  $A_{\mathfrak{p}} \hookrightarrow B_{\mathfrak{p}} := S^{-1}B$ , an integral extension (by the localization-

integral-extension theorem below), with  $A_{\mathfrak{p}}$  local with maximal ideal  $\mathfrak{p}A_{\mathfrak{p}}$ . By the previous lemma (applied to  $A_{\mathfrak{p}} \hookrightarrow B_{\mathfrak{p}}$  and the maximal ideal  $\mathfrak{p}A_{\mathfrak{p}}$ ), there is a maximal ideal  $\mathfrak{P}$  of  $B_{\mathfrak{p}}$  with  $\mathfrak{P} \cap A_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$ .

Let  $P \subseteq B$  be the contraction of  $\mathfrak{P}$  along  $B \rightarrow B_{\mathfrak{p}}$ . Then  $P \cap A = \mathfrak{p}$ , proving (i).

Moreover, the primes of  $B$  lying over  $\mathfrak{p}$  are in bijective correspondence with the maximal ideals of  $B_{\mathfrak{p}}$  lying over  $\mathfrak{p}A_{\mathfrak{p}}$  (via this same contraction/extension correspondence). Since distinct maximal ideals are never comparable by inclusion, neither are the corresponding primes of  $B$  lying over  $\mathfrak{p}$ , proving (ii).  $\square$

**Theorem 1.66.** Let  $A \hookrightarrow B$  be an extension of rings and  $C$  the integral closure of  $A$  in  $B$ . If  $S$  is a multiplicative subset of  $A$ , then  $S^{-1}C$  is the integral closure of  $S^{-1}A$  in  $S^{-1}B$ .

*Proof.* Let  $b/s \in S^{-1}B$  be integral over  $S^{-1}A$ :

$$\left(\frac{b}{s}\right)^n + \frac{a_1}{s_1} \left(\frac{b}{s}\right)^{n-1} + \cdots + \frac{a_n}{s_n} = 0.$$

Let  $t = s_1 \cdots s_n$ . Multiplying through by  $(st)^n$ ,

$$(st)^n \left[ \left(\frac{b}{s}\right)^n + \cdots \right] = 0$$

becomes a monic equation in  $bt$  with coefficients in  $A$ . Hence  $bt$  is integral over  $A$ , i.e.  $bt \in C$ , so

$$\frac{b}{s} = \frac{bt}{st} \in S^{-1}C.$$

$\square$

## CHAPTER 2

# PROBLEM DISCUSSIONS

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### 2.1 Assignment 0

*Convention.* By a ring, we mean a commutative ring with 1. Also, we exclude the zero ring.

1. Let  $I$  be an ideal of  $R$  and  $\text{rad}(I) := \{a \in R \mid a^n \in I \text{ for some } n \geq 1\}$ . Prove the following:

(a)  $\text{rad}(\text{rad}(I)) = \text{rad}(I)$

*Solution.* The containment  $\text{rad}(I) \subseteq \text{rad}(\text{rad}(I))$  is immediate, since  $I \subseteq \text{rad}(I)$  and  $\text{rad}(-)$  is monotone.

Conversely, let  $x \in \text{rad}(\text{rad}(I))$ . Then  $x^n \in \text{rad}(I)$  for some  $n$ , so  $(x^n)^m = x^{mn} \in I$  for some  $m$ . Hence  $x \in \text{rad}(I)$ .  $\square$

(b)  $\text{rad}(IJ) = \text{rad}(I \cap J) = \text{rad}(I) \cap \text{rad}(J)$

*Solution.* We show  $\text{rad}(I) \cap \text{rad}(J) \subseteq \text{rad}(IJ) \subseteq \text{rad}(I \cap J) \subseteq \text{rad}(I) \cap \text{rad}(J)$ , giving all three sets equal.

- $\text{rad}(I) \cap \text{rad}(J) \subseteq \text{rad}(IJ)$ : if  $x \in \text{rad}(I) \cap \text{rad}(J)$ , then  $x^n \in I$  and  $x^m \in J$  for some  $n, m$ , so  $x^{n+m} = x^n x^m \in IJ$ , i.e.  $x \in \text{rad}(IJ)$ .
- $\text{rad}(IJ) \subseteq \text{rad}(I \cap J)$ : since  $IJ \subseteq I \cap J$ , this follows from monotonicity of  $\text{rad}$ .
- $\text{rad}(I \cap J) \subseteq \text{rad}(I) \cap \text{rad}(J)$ : since  $I \cap J \subseteq I$  and  $I \cap J \subseteq J$ , monotonicity gives  $\text{rad}(I \cap J) \subseteq \text{rad}(I)$  and  $\text{rad}(I \cap J) \subseteq \text{rad}(J)$ .

$\square$

(c)  $\text{rad}(I) = R$  if and only if  $I = R$

*Solution.*  $(\Rightarrow)$  If  $\text{rad}(I) = R$ , then for every  $a \in R$  there is  $n \in \mathbb{N}$  with  $a^n \in I$ . Taking  $a = 1$ :  $1^n = 1 \in I$ , so  $I = R$ .

$(\Leftarrow)$  Trivial: if  $I = R$  then  $\text{rad}(I) = R$  as well.  $\square$

(d)  $\text{rad}(I + J) = \text{rad}(\text{rad}(I) + \text{rad}(J))$

*Solution.* ( $\subseteq$ ) Let  $x \in \text{rad}(I + J)$ . Then  $x^k = u + v$  for some  $k \in \mathbb{N}$ ,  $u \in I$ ,  $v \in J$ . Since  $u \in I \subseteq \text{rad}(I)$  and  $v \in J \subseteq \text{rad}(J)$ ,  $x^k \in \text{rad}(I) + \text{rad}(J)$ , so  $x \in \text{rad}(\text{rad}(I) + \text{rad}(J))$ .

( $\supseteq$ ) Since  $I \subseteq I + J$ , monotonicity gives  $\text{rad}(I) \subseteq \text{rad}(I + J)$ ; similarly  $\text{rad}(J) \subseteq \text{rad}(I + J)$ . As  $\text{rad}(I + J)$  is an ideal,  $\text{rad}(I) + \text{rad}(J) \subseteq \text{rad}(I + J)$ . Applying  $\text{rad}(-)$  and using part (a),

$$\text{rad}(\text{rad}(I) + \text{rad}(J)) \subseteq \text{rad}(\text{rad}(I + J)) = \text{rad}(I + J).$$

□

(e) If  $P \in \text{Spec}(R)$ , then  $\text{rad}(P^n) = P$  for all  $n > 0$

*Solution.* First,  $\text{rad}(P) = P$ : the containment  $P \subseteq \text{rad}(P)$  always holds, and  $\text{rad}(P) \subseteq P$  follows since  $P$  is prime (if  $x^n \in P$ , primality forces  $x \in P$ ).

Now induct on  $n$ . The base case  $n = 1$  is  $\text{rad}(P) = P$ , shown above. For the inductive step, using part (b) with  $I = P^{n-1}$ ,  $J = P$ :

$$\text{rad}(P^n) = \text{rad}(P^{n-1} \cdot P) = \text{rad}(P^{n-1}) \cap \text{rad}(P) = P \cap P = P,$$

using the induction hypothesis  $\text{rad}(P^{n-1}) = P$ .

□

2. Let  $R[X]$  be the polynomial ring over a ring  $R$ . Prove that the nilradical of  $R[X]$  is equal to the Jacobson radical of  $R[X]$ .

*Solution.* Since every maximal ideal is prime, intersecting over the (larger) family of prime ideals gives a smaller or equal intersection than intersecting over maximal ideals alone; hence, for any ring,  $\mathcal{N}(R[X]) \subseteq J(R[X])$ .

For the reverse containment, we use the identity  $\mathcal{N}(R[X]) = (\mathcal{N}(R))[X]$  (proof deferred). Take  $f(X) \in J(R[X])$ . By the standard characterization of the Jacobson radical,  $1 - g(X)f(X)$  is a unit for every  $g(X) \in R[X]$ ; taking  $g(X) = -X$ ,

$$1 + Xf(X) \text{ is a unit in } R[X].$$

By the Lecture 1 proposition on units of  $R[X]$ , since the constant term of  $1 + Xf(X)$  is 1 (a unit), all remaining coefficients of  $1 + Xf(X)$  – i.e. all coefficients of  $f(X)$  – are nilpotent. Hence  $f(X) \in (\mathcal{N}(R))[X] = \mathcal{N}(R[X])$ .

Therefore  $J(R[X]) \subseteq \mathcal{N}(R[X])$ , and combined with the reverse containment above,  $\mathcal{N}(R[X]) = J(R[X])$ . □

3. Let  $R[X]$  be the polynomial ring over a ring  $R$ . Let  $J \subset R$  be an ideal. Define

$$J[X] := \{a_0 + a_1X + \cdots + a_nX^n \mid a_i \in J \forall i\}.$$

Prove that  $J[X]$  is an ideal of  $R[X]$ . Give an example of an ideal of  $R[X]$  which is not of this form. Prove that the rings  $R[X]/J[X]$  and  $(R/J)[X]$  are isomorphic.

*Solution.* That  $J[X]$  is an ideal of  $R[X]$  is a direct verification (closed under addition and absorbs multiplication by  $R[X]$ , coefficientwise, since  $J$  is an ideal of  $R$ ).

**Example of an ideal not of this form.** Let

$$I = \{p(X) \in R[X] \mid \exists q(X) \in R[X] \text{ such that } p(X) = X \cdot q(X)\} = \langle X \rangle,$$

the ideal of polynomials with zero constant term. This is not of the form  $J[X]$  for any proper ideal  $J \subsetneq R$ : an element of  $J[X]$  has *every* coefficient in  $J$ , whereas  $I$  places no constraint on coefficients beyond the constant term.

**Isomorphism.** Define  $\varphi : R[X]/J[X] \rightarrow (R/J)[X]$  by

$$\varphi(r(X) + J[X]) = \sum_{i=0}^{\deg r} (r_i + J) X^i,$$

i.e. reduce each coefficient of  $r(X)$  modulo  $J$ . This is induced by the coefficientwise reduction map  $R[X] \rightarrow (R/J)[X]$ , which is a surjective ring morphism with kernel exactly  $J[X]$ ; by the First Isomorphism Theorem,  $\varphi$  is a well-defined ring isomorphism.  $\square$

4. A commutative ring is called a *local ring* if it has only one maximal ideal. Let  $R$  be a local ring with unique maximal ideal  $\mathfrak{m}$ . Prove that any element  $a \in R \setminus \mathfrak{m}$  is a unit in  $R$ . Conversely, let  $R$  be a ring such that the set of non-units forms an ideal  $M$ . Prove that  $R$  is local with maximal ideal  $M$ .

*Solution.* ( $\Rightarrow$ ) FTSOC, suppose  $a \in R \setminus \mathfrak{m}$  is not a unit. Then  $\langle a \rangle \neq R$ , i.e.  $1 \notin \langle a \rangle$ . Let

$$A = \{I \subseteq R \text{ an ideal} \mid a \in I \text{ but } 1 \notin I\}.$$

Since  $\langle a \rangle \in A$ ,  $A \neq \emptyset$ . By Zorn's Lemma, let  $M$  be a maximal element of  $A$  (ordered by inclusion). One checks  $M$  is a maximal ideal of  $R$ . Since  $R$  is local,  $M = \mathfrak{m}$ , and since  $a \in M$ , we get  $a \in \mathfrak{m}$  – contradicting  $a \notin \mathfrak{m}$ . Hence  $a$  must be a unit.

**(Converse)** Suppose  $M = \{x \in R \mid x \text{ is not a unit}\}$  is an ideal. Every proper ideal  $I$  of  $R$  consists entirely of non-units (else  $I$  would contain a unit, forcing  $I = R$ ), so  $I \subseteq M$ . In particular  $M$  itself is a proper ideal (as  $1 \notin M$ ), and since every proper ideal is contained in  $M$ ,  $M$  is the unique maximal ideal of  $R$ . Hence  $R$  is local with maximal ideal  $M$ .  $\square$

5. An element  $e \in R$  is called an *idempotent* if  $e^2 = e$ . Let  $R$  be a local ring. Prove that the only idempotents of  $R$  are 0 and 1.

*Solution.* **Lemma.** If  $R$  is a local ring, then for every  $a \in R$ , either  $a$  or  $1 - a$  is a unit.

*Proof.* Suppose neither  $a$  nor  $1 - a$  is a unit. Then  $a \in \mathfrak{m}$  and  $1 - a \in \mathfrak{m}$  (the unique maximal ideal), so  $1 = a + (1 - a) \in \mathfrak{m}$ , forcing  $\mathfrak{m} = R$  – a contradiction.

Now let  $e \in R$  be idempotent:  $e^2 = e$ , i.e.  $e(e - 1) = 0$ . By the lemma, at least one of  $e, 1 - e$  is a unit; multiplying  $e(e - 1) = 0$  by its inverse forces the other factor to vanish. Hence  $e = 0$  or  $e = 1$ .  $\square$

6. Let  $R = \mathbb{Z}[i]$ . Prove that the ideal  $P = \langle 1 + i \rangle$  is a prime ideal of  $R$ . Prove that  $\langle 2 \rangle = P^2$  in  $R$ .
7. Prove that the ring  $\mathbb{Z}[i]/\langle 2 + i \rangle$  is isomorphic to the ring  $\mathbb{Z}/5\mathbb{Z}$ .
8. Let  $R$  be a ring and  $a, b \in R$  be two *comaximal* elements, i.e., there exist  $x, y \in R$  such that  $xa + yb = 1$ . Prove that for any  $m, n \in \mathbb{N}$ , one has  $a^m$  and  $b^n$  are also comaximal (i.e., there exist  $\lambda, \mu \in R$  such that  $\lambda a^m + \mu b^n = 1$ ).
9. Let  $R$  be an integral domain. Let  $a, b \in R$  be such that their gcd exists. If  $d, d'$  are both gcds of  $a, b$ , prove that  $d$  and  $d'$  are associates.
10. Let  $R$  be an integral domain such that  $\gcd(a, b)$  exists for any  $a, b \in R \setminus \{0\}$ . Then prove that  $\text{lcm}(a, b)$  exists for any  $a, b \in R \setminus \{0\}$ . (Hint: First prove that  $\gcd(ma, mb) = m \gcd(a, b)$ .)
11. Let  $R$  be an integral domain such that  $\text{lcm}(a, b)$  exists for any  $a, b \in R \setminus \{0\}$ . Then prove that  $\gcd(a, b)$  exists for any  $a, b \in R \setminus \{0\}$ .
12. Let  $R$  be an integral domain such that  $\gcd(a, b)$  exists for any  $a, b \in R \setminus \{0\}$ . If  $\gcd(a, b) = 1$  and  $\gcd(a, c) = 1$ , then prove that  $\gcd(a, bc) = 1$ .
13. Let  $R = \mathbb{Z}[\sqrt{-5}]$ . Verify the following:

(a) There are two distinct (i.e. non-associate) factorizations of 6 into irreducibles in  $R$ :

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}).$$

(b) The ideals  $\langle 2, 1 \pm \sqrt{-5} \rangle, \langle 3, 1 \pm \sqrt{-5} \rangle$  are all prime ideals.

(c) The following factorization of ideals into prime ideals holds:

- $\langle 2 \rangle = \langle 2, 1 + \sqrt{-5} \rangle^2$ ,
- $\langle 3 \rangle = \langle 3, 1 + \sqrt{-5} \rangle \langle 3, 1 - \sqrt{-5} \rangle$ ,
- $\langle 1 + \sqrt{-5} \rangle = \langle 2, 1 + \sqrt{-5} \rangle \langle 3, 1 + \sqrt{-5} \rangle$ ,
- $\langle 1 - \sqrt{-5} \rangle = \langle 2, 1 + \sqrt{-5} \rangle \langle 3, 1 - \sqrt{-5} \rangle$ .

(d) Uniqueness of factorization is restored in (a) at the level of ideals.

(e) The ideal  $\langle 2, 1 + \sqrt{-5} \rangle$  is not a principal ideal.

14. Show that  $\langle 2, 1 + \sqrt{-17} \rangle$  is not a principal ideal in  $\mathbb{Z}[\sqrt{-17}]$ . Is this a prime ideal?

15. Let  $R$  be an ID. Assume that:

(a) any two  $a, b \in R \setminus \{0\}$  have gcd, say,  $d$ , and there are  $x, y \in R$  such that  $d = ax + by$ ;

- (b) if  $a_1, a_2, \dots$  are nonzero elements of  $R$  such that  $a_{i+1}$  divides  $a_i$  for  $i = 1, 2, \dots$ , then there is  $N > 0$  such that  $a_n = u_n a_N$  for some unit  $u_n \in R$  for all  $n \geq N$ .

Show that  $R$  is a PID.

16. Prove that  $R[X]/\langle X^3 + X \rangle$  is a product of fields.
17. Let  $m, n \in \mathbb{Z}$  be two nonzero elements and  $d = \gcd(m, n)$  in  $\mathbb{Z}$ . Prove that  $d$  is also a gcd of  $m, n$  in  $\mathbb{Z}[i]$ .
18. Let  $m, n \in \mathbb{Z}$  be two nonzero elements and let  $m$  divide  $n$  in  $\mathbb{Z}[i]$ . Does  $m$  also divide  $n$  in  $\mathbb{Z}$ ?
19. Let  $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$  be an exact sequence of  $R$ -modules. Let  $M', M''$  be both finitely generated. Show that  $M$  is finitely generated.
20. From Dummit–Foote: Exercise 10.1: 5, 8, 9, 10, 11, 15.
21. From Dummit–Foote: Exercise 10.2: 3, 4, 5, 6, 7, 9, 10, 11, 12, 13.
22. An  $R$ -module is called *simple* if it is not the zero module and it has no proper submodule.
  - (a) Prove that an  $R$ -module  $M$  is simple if and only if it is isomorphic (as an  $R$ -module) to  $R/\mathfrak{m}$ , where  $\mathfrak{m}$  is a maximal ideal of  $R$ . Show also that  $M$  is generated by any non-zero element.
  - (b) Let  $M, N$  be simple  $R$ -modules. Prove that a module morphism  $\phi : M \rightarrow N$  is either the zero morphism or an isomorphism. Conclude that if  $M$  is simple,  $\text{Hom}_R(M, M)$  is a division ring.
23. Let  $M$  be an additive abelian group and  $\text{End}(M)$  be the set of all group morphisms from  $M$  to  $M$ .
  - (a) Show that  $\text{End}(M)$  is a ring with identity (where addition is defined pointwise, and multiplication is composition of maps).
  - (b) Show that  $M$  is an  $\text{End}(M)$ -module under the action:
 
$$(f, m) \mapsto fm := f(m).$$
  - (c) Let  $R$  be a ring and  $\phi : R \rightarrow \text{End}(M)$  be a ring morphism. Show that  $M$  is an  $R$ -module under the action:
 
$$(r, m) \mapsto rm := \phi(r)(m).$$
24. (Converse of 4(c)): Let  $R$  be a ring and  $M$  be an  $R$ -module. Show that there is a ring morphism  $\phi : R \rightarrow \text{End}(M)$ .
25. Let  $M$  be an  $R$ -module and  $A, B, C$  be its submodules such that

$$A \subseteq B, \quad A + C = B + C, \quad A \cap C = B \cap C.$$

Prove that  $A = B$ .

26. Let  $f : M \rightarrow N$  be an  $R$ -module morphism. Let  $A$  be a submodule of  $M$ . Prove that  $f^{-1}(f(A)) = A + \text{Ker}(f)$ . Now let  $B$  be a submodule of  $N$ . Prove that  $f(f^{-1}(B)) = B \cap \text{Im}(f)$ . Also, show that

$$f(A \cap f^{-1}(B)) = f(A) \cap B.$$

27. Let  $R$  be a commutative ring. Show that a map  $f : R \times R \rightarrow R$  is an  $R$ -module morphism if and only if  $f(x, y) = \alpha x + \beta y$  for some  $\alpha, \beta \in R$ .
28. Let  $f : M \rightarrow N$  be an  $R$ -module morphism. Show that  $f$  can be expressed as the composition of three  $R$ -module morphisms (for appropriate  $R$ -modules):  $f = g \circ h \circ k$ , where  $g$  is surjective,  $h$  is an isomorphism, and  $k$  is injective.
29. Let  $R$  be an integral domain and  $a \in R$  be non-zero. Let  $n$  be any positive integer. Prove that  $Ra^n / Ra^{n+1}$  is isomorphic to  $R/Ra$  (as  $R$ -modules).



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